4190.428 Introduction to Machine Learning

Regression

Jaesik Park
CSE & IPAI, SNU
jaesik.park@snu.ac.kr

Overview

- Regression : a slightly different form of machine-learning problem
  - It is still supervised learning, so our data will still have the form :

$$S_n = \left\{ (\mathbf{x}^{(1)}, y^{(1)}), \ldots, (\mathbf{x}^{(n)}, y^{(n)}) \right\}$$

- But now, instead of the y values being discrete, they will be real-valued, and so our hypotheses will have the form

$$h : \mathbb{R}^d \to \mathbb{R}$$

- This is a good framework when we want to predict a numerical quantity, rather than to divide the inputs into categories.

Regression

- The first step is to pick a loss function, to describe how to evaluate the quality of the predictions our hypothesis is making, when compared to the "target" y values in the data set.
- The choice of loss function is part of modeling your domain.
- In the absence of additional information about a regression problem, we typically use **squared error (SE)** :

$$\text{Loss}(\text{guess}, \text{actual}) = (\text{guess} - \text{actual})^2$$

  - It penalizes guesses that are too high the same amount as it penalizes guesses that are too low, and has a good mathematical justification in the case that your data are geared from an underlying hypothesis, but with Gaussian-distributed noise added to the y values.

Regression

- We will consider the case of a linear hypothesis class :

$$h(\mathbf{x}; \theta, \theta_0) = \theta^\top \mathbf{x} + \theta_0$$

- Remembering that we can get a rich class of hypotheses by performing a non-linear feature transformation before doing the regression.
  - So, $\theta^\top \mathbf{x} + \theta_0$ is a linear function of x, but $\theta^\top \phi(\mathbf{x}) + \theta_0$ is a non-linear function of x is ϕ is a non-linear function of x.

Regression

- We will treat regression as an optimization problem, in which, given a data set $\mathscr{D}$, we wish to find a linear hypothesis that minimizes mean squared error.
- Our objective, often called **mean squared error**, is to find values for $\boldsymbol{\Theta} = (\theta, \theta_0)$ that minimize

$$J(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} + \theta_0 - y^{(i)} \right)^2,$$

- Resulting in the solution :

$$\theta^*, \theta_0^* = \arg\min_{\theta, \theta_0} J(\theta, \theta_0),$$

Analytical solution: ordinary least squares

- Our very interesting aspect of the problem finding a linear hypothesis that minimizes mean squared error (this general problem is often called **ordinary least squared (OLS))** is that we can find a closed-form formula for the answer.
  - Everything is easier to deal with if we assume that the $x^{(i)}$ have been augmented with an extra input dimension (feature) that always has value 1 , so may ignore θ_0.
  - We will approach this just like minimization problem from calculus homework: take the derivative of J with respect to θ, set it to zero, and solve for θ.
  - There is an additional step required, to check that the resulting θ is a minimum but we won't work through that here.

Analytical solution: ordinary least squares

- It is possible to approach this problem by :
  - Finding $\frac{\partial J}{\partial \theta_k}$, for $k \in 1, \ldots, d$
  - Constructing a set of k equations of the form $\frac{\partial J}{\partial \theta_k} = 0$, and
  - Solving the system for values of θ_k
  - That works just fine.

- To get practice for applying techniques like this to more complex problems, we will work through a more compact matrix view.

Analytical solution: ordinary least squares

- We can think of our training data in terms of matrices X and Y, where each column of X is an example, and each "column" of Y is the corresponding target output value:

$$X = \begin{bmatrix} x_1^{(1)} & \cdots & x_1^{(n)} \\ \vdots & \ddots & \vdots \\ x_d^{(1)} & \cdots & x_d^{(n)} \end{bmatrix}, \quad Y = \begin{bmatrix} y^{(1)} & \cdots & y^{(n)} \end{bmatrix}.$$

- **Study question**: What are the dimensions of X and Y?

Analytical solution: ordinary least squares

- It most textbooks, they think of an individual example $x^{(i)}$ as a row, rather than a column.
- So that we get as answer that will be recognizable to you, we are going to define a new matrix and vector, W and T, which are just transposes of our X and Y, and then work with them:

$$W = X^\top = \begin{bmatrix} x_1^{(1)} & \cdots & x_d^{(1)} \\ \vdots & \ddots & \vdots \\ x_1^{(n)} & \cdots & x_d^{(n)} \end{bmatrix}, \quad T = Y^\top = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{bmatrix}.$$

- **Study Question**: What are the dimensions of W and T?
- Now we can write:

$$J(\theta) = \frac{1}{n} \underbrace{\left(W\theta - T\right)^\top}_{1\times n} \underbrace{\left(W\theta - T\right)}_{n\times 1} = \frac{1}{n} \sum_{i=1}^{n} \left(\left(\sum_{j=1}^{d} w_{ij}\theta_j \right) - T_i \right)^2$$

Analytical solution: ordinary least squares

- And using facts about matrix/vector calculus, we get

$$\nabla_\theta J = \frac{2}{n} \underbrace{W^\top}_{d \times n} \underbrace{\left(W\theta - T\right)}_{n \times 1}.$$

- Setting to 0 and solving, we get [Try yourself]:

$$\frac{2}{n} W^\top \left(W\theta - T\right) = 0$$

$$W^\top W\theta - W^\top T = 0$$

$$W^\top W\theta = W^\top T$$

$$\theta = \left(W^\top W\right)^{-1} W^\top T$$

Analytical solution: ordinary least squares

- And the dimension work out.

$$\theta = \underbrace{\left(W^\top W\right)^{-1}}_{d \times d} \underbrace{W^\top}_{d \times n} \underbrace{T}_{n \times 1} .$$

- So, given our data, we can directly compute the linear regression that minimizes mean squared error.

Analytical solution: ordinary least squares

- Well, actually, there are some kinds of trouble we can get into.
- What if $(W^\top W)$ is not invertible?

- **Study Question**:
  - Consider, for example, a situation where the data-set is just the same point repeated twice: $x^{(1)} = x^{(2)} = (1,2)^\top$.
  - What is W in the case? , What is $W^\top W$?, What is $(W^\top W)^{-1}$?.

- Another kind of problem is **overfitting**: we have formulated an objective that is just about fitting the data as well as possible, but as we discussed in the context of margin maximization, we might also want to regularize to keep the hypothesis from getting too attached to the data.

Regularizing linear regression

- We address both the problem of not being able to invert $(W^\top W)^{-1}$ and the problem of overfitting using a mechanism called **ridge regression**.
- We add a regularization term $||\theta||^2$ to the OLS objective, with trade-off parameter λ.
  - **Study Question**: When we add a regularizer of the form $||\theta||^2$, what is our most "preferred" value of θ, in the absence of any data?

- Here is the ridge regression objective function:

$$J_{\text{ridge}}(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} + \theta_0 - y^{(i)}\right)^2 + \lambda ||\theta||^2$$

Regularizing linear regression

- Larger λ values pressure θ values to be near zero.

- Note that we don't penalize θ_0; intuitively, θ_0 is what "floats" the regression surface to the right level for the data you have, and so you shouldn't make it harder to fit a data set where the y values tent to be around one million than one where they tend to be around one.

- The other parameters control the orientation of the regression surface, and we prefer it to have a not-too-crazy orientation.

Regularizing linear regression

ADVANCED

- There is an analytical expression for the θ, θ_0 values that minimize J_{ridge}, but it's a little bit more complicated to derive than the solution for OLS because θ_0 needs special treatment.
- If we decide not to treat θ_0 specially, then we get:

$$\nabla_\theta J_{\text{ridge}} = \frac{2}{n} W^\top (W\theta - T) + 2\lambda\theta$$

Regularizing linear regression

ADVANCED

- Setting to 0 and solving we get [Try yourself]:

$$\frac{2}{n}W^\top(W\theta - T) + 2\lambda\theta = 0$$

$$\frac{1}{n}W^\top W\theta - \frac{1}{n}W^\top T + \lambda\theta = 0$$

$$\frac{1}{n}W^\top W\theta + \lambda\theta = \frac{1}{n}W^\top T$$

$$W^\top W\theta + n\lambda\theta = W^\top T$$

$$(W^\top W + n\lambda I)\theta = W^\top T$$

$$\theta = (W^\top W + n\lambda I)^{-1}W^\top T$$

- So, $\theta_{\text{ridge}} = (W^\top W + n\lambda I)^{-1}W^\top T$ which becomes invertible when $\lambda > 0$.
  - **Study Question**: Derive this version of the ridge regression solution.

Regularizing linear regression

- **Talking about regularization**: In machine learning in general, not just regression, it is useful to distinguish two ways in which a hypothesis $h \in \mathscr{H}$ might contribute to errors on test data. We have
  - **Structural error**: This is error that arises because there is no hypothesis $h \in \mathscr{H}$ that will perform well on the data, for example because the data was really generated by a sine wave but we are trying to fit it with a line.
  - **Estimation error**: This is error that arises because we do not have enough data to allow us to choose a good $h \in \mathscr{H}$.

Regularizing linear regression

- When we increase λ, we tend to increase structural error but decrease estimation error, and vice versa.
- **Study Question**: Consider using a polynomial basis of order k as a feature transformation ϕ on your data. Would increasing k tend to increase or decrease structural error? What about estimation error?

Regularizing linear regression

- Inverting the $d \times d$ matrix ($W^\top W$) takes $O(d^3)$ time, which makes the analytic solution impractical for large d.
  - **Study Question** : Why is having large n not as much of a computational problem as having large d?
- Therefore, If we have high-dimensional data, we can fall back on gradient descent

Optimization via gradient descent

- Recall the ridge objective

$$J_{\text{ridge}}(\theta, \theta_0) = \frac{1}{n} \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} + \theta_0 - y^{(i)}\right)^2 + \lambda \|\theta\|^2$$

- And its gradient with respect to θ

$$\nabla_\theta J = \frac{2}{n} \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} + \theta_0 - y^{(i)}\right) \mathbf{x}^{(i)} + 2\lambda\theta$$

Optimization via gradient descent

- And partial derivative with respect to θ_0

$$\frac{\partial J}{\partial \theta_0} = \frac{2}{n} \sum_{i=1}^{n} \left(\theta^\top \mathbf{x}^{(i)} + \theta_0 - y^{(i)} \right)$$

- Even better, the objective functions for OLS and ridge regression are **convex**, which means they have only one minimum, which means, with a small enough step size, gradient descent is **guaranteed** to find the optimum. (Gradient Descent Convergence Theorem)

QnA

- Any questions?
- Note: this lecture notes are largely adopted from
  - MIT 6.036(2019 Fall), Stanford CS229(2023-2024 Summer)